

**DATE:**

**EXPT NO:04**

## **EDA– Data Inspection, Filtering**

### **AIM:**

To visualize the Earthquake dataset using Python and create different charts to understand earthquake patterns, magnitude distributions, and identify important geographic features.

### **PROCEDURE:**

1. Start the program.
2. Import Pandas, Matplotlib, and Seaborn libraries.
3. Load the Earthquake CSV dataset.
4. Display the first few records using head().
5. Create a bar chart to compare seismic events based on magnitude.
6. Create a box plot to show the distribution of earthquake magnitudes.
7. Create a line chart to analyze seismic trends across recorded events.
8. Add suitable titles, axis labels, and legends.
9. Display all the visualizations.
10. Analyze the graphs and identify important patterns.
11. Stop the program.

### **CODE:**

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("earthquake_data.csv")

# Select numerical columns
num_cols = df.select_dtypes(include="number").columns
print("Numerical columns:", num_cols.tolist())

# BAR CHART
plt.figure(figsize=(8, 5))
df[num_cols[0]].value_counts().head(10).plot(kind="bar")
plt.title("Bar Chart")
plt.xlabel(num_cols[0])
plt.ylabel("Count")
plt.show()

# BOX PLOT
```

```
plt.figure(figsize=(8, 5))
sns.boxplot(x=df[num_cols[0]])
plt.title("Box Plot")
plt.xlabel(num_cols[0])
plt.show()

# LINE CHART
plt.figure(figsize=(8, 5))
plt.plot(df[num_cols[0]].head(20), marker="o")
plt.title("Line Chart")
plt.xlabel("Earthquake Event")
plt.ylabel(num_cols[0])
plt.grid()
plt.show()
```

**OUTPUT:**

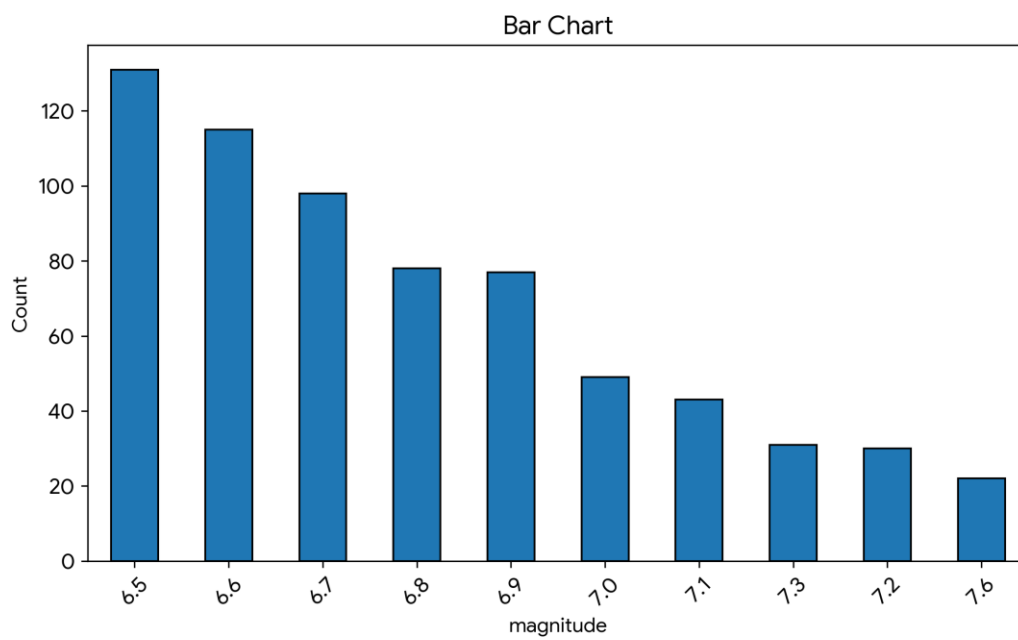
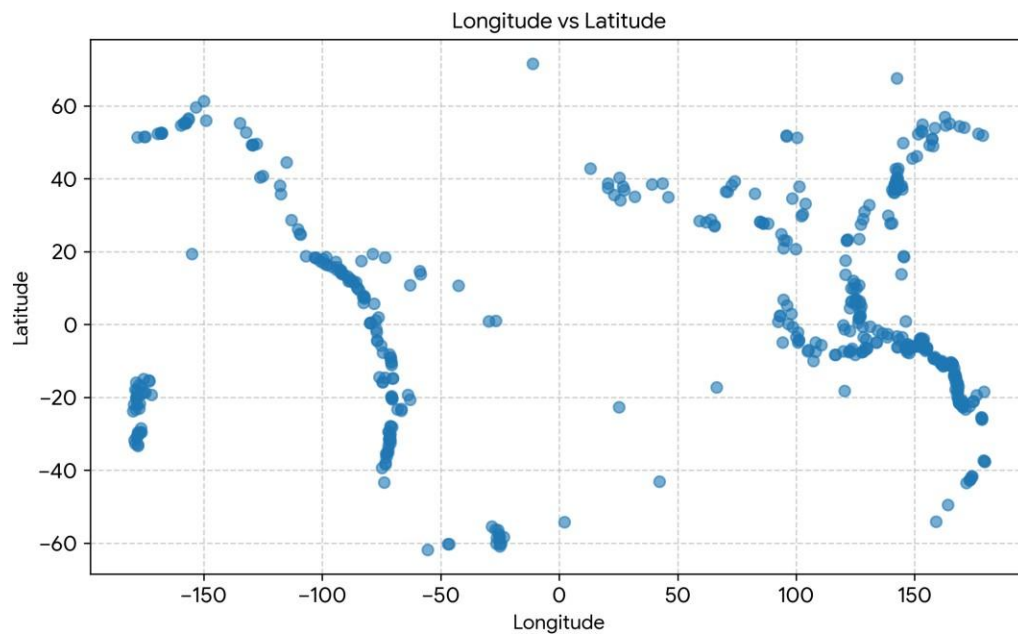
Columns:

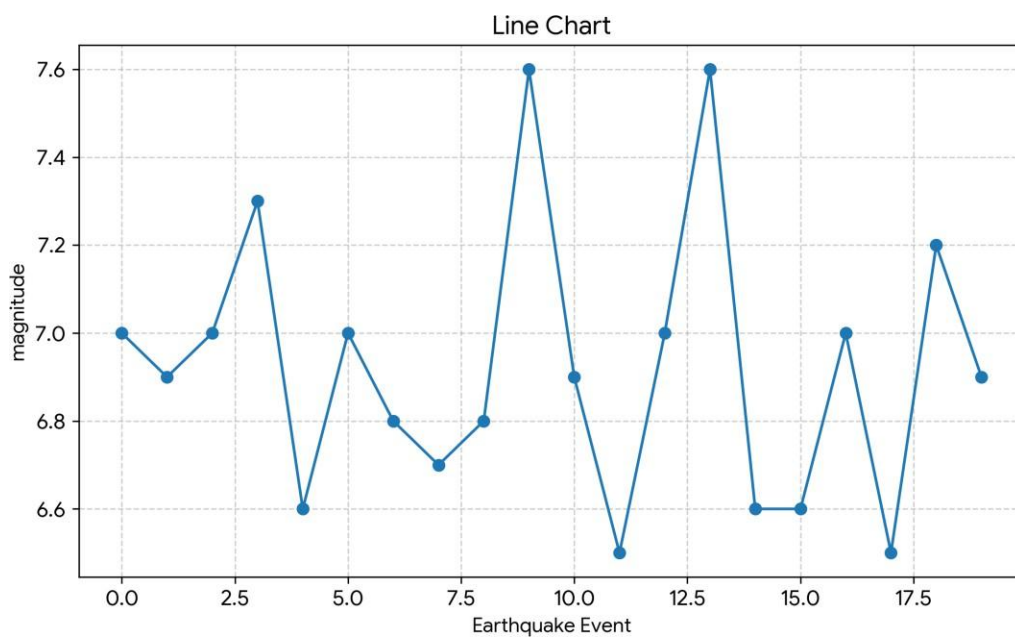
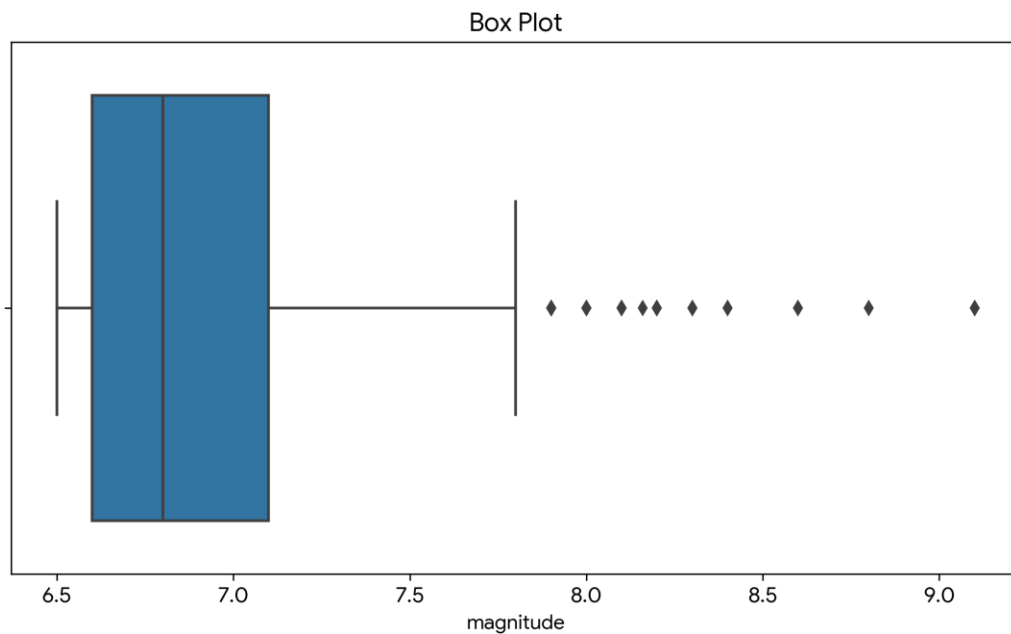
```
['title', 'magnitude', 'date_time', 'cdi', 'mmi', 'alert', 'tsunami', 'sig',
'net', 'nst', 'dmin', 'gap', 'magType', 'depth', 'latitude', 'longitude',
'location', 'continent', 'country']
```

Numerical columns: ['magnitude', 'cdi', 'mmi', 'tsunami', 'sig', 'nst', 'dmin', 'gap', 'depth', 'latitude', 'longitude']

First 5 Records:

[illegible]





**RESULT:**